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Palladium(0)-catalyzed synthesis of cyclic glucosides†

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A highly regio- and stereo-selective synthesis of cyclic β-D-glucosides **3** *via* Pd(0)-catalyzed coupling cyclization of allenyl β-D-glucoside **1** and organic iodides in 20–38% yields is reported. After the deacetylation of **3**, we obtained cyclic β-D-glucoses **4** in 90–97% yields, which may have SGLT2 inhibition activities.

Nowadays, the number of people suffering from type 2, non-insulin dependent, diabetes mellitus (NIDDMa) is increasing.[1] Since the first sodium glucose co-transporter 2 (SGLT2) inhibitor Phlorizin,[2] a naturally-occurring *O*-aryl glycoside was discovered, more and more similar derivatives[3] have been developed by chemists. Usually the skeletons of SGLT2 inhibitors are *C*- or *O*-glycosides bearing more than two aromatic or heteroaromatic rings (Fig. 1).[3] However, there are limited reports on the synthesis of cyclic glycosides with SGLT2 inhibitors.[4]

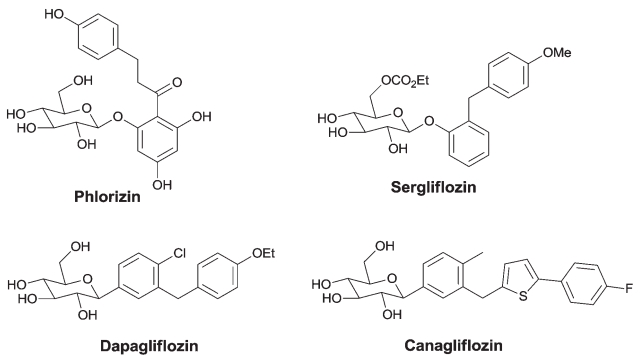
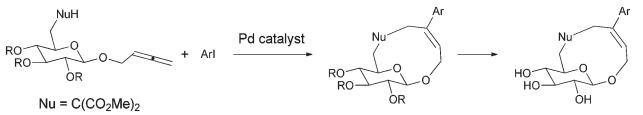


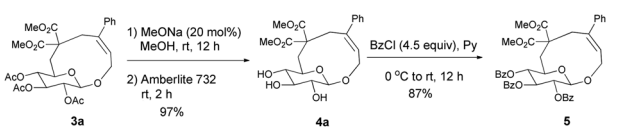
Fig. 1 Typical structures of some *C*-aryl or *O*-aryl glucoside SGLT2 inhibitors.

Recently, several reports on the Pd-catalyzed synthesis of medium-sized rings or macrocycles *via* allenes with a nucleophilic functionality and organic halides have been reported.[5] We reasoned that this method could be applied to the construction of aryl or heteroaryl substituted cyclic glycosides, which may also have inhibition activities to type 2 diabetes mellitus (Scheme 1). Usually medium sized rings are very hard to prepare due to entropic/enthalpic factors, especially here where an oxygen bridge exists in the designed cyclic glucosides.[6]



Scheme 1 Design of new cyclic glucoside SGLT2 inhibitors.

Initially, we conducted the reaction of allenyl β-D-glucoside **1**[7] with PhI (2.0 equiv.) under the catalysis of $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%), TFP (10 mol%, TFP = tris(2-furyl)phosphine) and K_2CO_3 (4.0 equiv.) at 60 °C. After screening a series of solvents, we found that the reaction using DMF afforded cyclic glucoside **3a** in the highest NMR yield of 21% (entries 1–4, Table 1). The skeleton of the cyclic glucosides on the ring was further established by the X-ray diffraction study of compound **5** (Fig. 2), which was synthesized by deacetylation and benzoylation of **3a** [eqn (1)].[8] Increasing the temperature to 80 °C could shorten the reaction time to 5 h with a similar yield (entry 5, Table 1). The yield could be increased to 29% when the temperature is increased to 100 °C (entry 6, Table 1). Further reaction at 120 °C gave a lower yield (entry 7, Table 1).



(1)

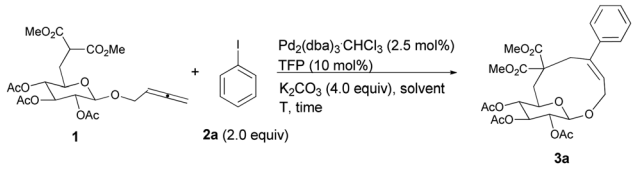
It is worth mentioning that the same yield of cyclic glucoside may be obtained with 2.0 equiv. of K_2CO_3 (entry 1, Table 2). We then conducted the reaction under diluted conditions and the yield improved to 37% when the concentration of **1** was decreased to 0.025 M (entries 1–5, Table 2). Also the

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†Electronic supplementary information (ESI) available: Experimental section, characterization of all compounds, and copies of ^1H and ^{13}C NMR spectra of selected compounds. CCDC 987494. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c4ob01975j

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Table 1 Effects of solvent and temperature[a]



Entry	Solvent	T (°C)	Time (h)	NMR yield of **3a**[b] (%)
1	DMA	60	15	17
2	Dioxane	60	40	0[c]
3	DMSO	60	11	14
4	DMF	60	14	21
5	DMF	80	5	19
6	DMF	100	5	29
7	DMF	120	5	19

[a] The reaction was conducted with a concentration of **1** of 0.067 M. [b] Determined by ^{1}H NMR analysis of the crude products with mesitylene as the internal standard. [c] 83% of **1** was recovered.

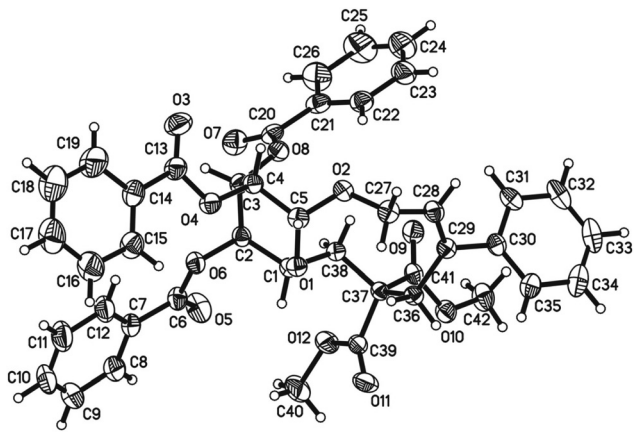
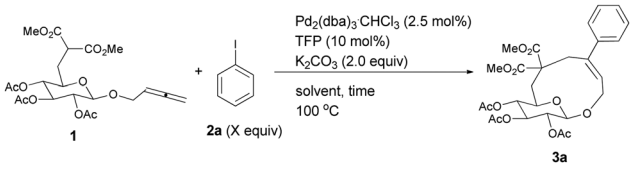


Fig. 2 ORTEP representation of compound **5**.

amount of **2a** could be lowered to 1.2 equiv. with a 38% yield of **3a** (entries 6 and 7, Table 2). It was found that a mixture of DMF–H_2O (80/1) is more effective with 39% NMR yield (entry 8, Table 2). Increasing the loading of water led to poorer results (entries 9 and 10). In addition, replacing TFP with PPh_3 afforded lower yield of **3a** (entry 11, Table 2).

Thus, the following [$Pd_2(dba)_3$·$CHCl_3$ (2.5 mol%), TFP (10 mol%), K_2CO_3 (2.0 equiv.) and **2** (1.2 equiv.) in DMA, 100 °C] were established as the optimized conditions for further study with the typical results summarized in Table 3. The following issues should be noted: (i) yields of cyclic glucosides **3** range from 20% to 38% while the reactions were carried out on a 1.0 mmol scale. This is quite remarkable since the formation of ten-membered cyclic glucosides is not easy. (ii) Not only electron-rich (entries 2 and 3, Table 3) but also electron-deficient (entries 4–10, Table 3) substituents could be introduced into aryl iodides. (iii) Also thienyl iodide (**2j**) could react with **1**, which afforded **3j** in 34% yield (entry 11, Table 3). (iv) Vinyl iodide (**2k**) could also be applied in this

Table 2 Effects of concentration of **1**, amount of **2a**, H_2O, and ligand.[a]

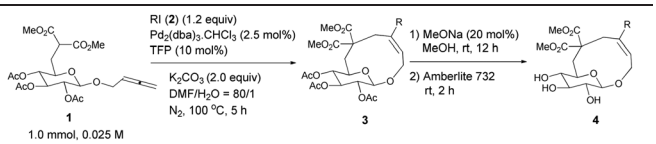


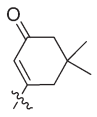
Entry	X (equiv.)	Solvent	Conc. of **1** (mol L^{-1})	Time (h)	NMR yield of **3a**[b] (%)
1	2.0	DMF	0.067	5	29
2	2.0	DMF	0.050	4	26
3	2.0	DMF	0.033	4	31
4	2.0	DMF	0.025	8	37
5	2.0	DMF	0.014	8	37
6	1.5	DMF	0.025	7	37
7	1.2	DMF	0.025	7	38
8	1.2	DMF–H_2O (80/1)	0.025	3	39
9	1.2	DMF–H_2O (40/1)	0.025	3	37
10	1.2	DMF–H_2O (10/1)	0.025	3	0
11[c]	1.2	DMF–H_2O (80/1)	0.025	20	28

[a] Reaction conditions: **1** (0.1 mmol), $Pd_2(dba)_3$.$CHCl_3$ (2.5 mol%), TFP (10 mol%), K_2CO_3 (2.0 equiv.), 100 °C. [b] Determined by ^{1}H NMR analysis of the crude products with mesitylene as the internal standard. [c] 10 mol% of PPh_3 was used instead of TFP.

reaction affording the corresponding product **3k** in 20% isolated yield (entry 12, Table 3). (v) We also investigated the reaction of **1** and **2f** on a 4 mmol scale under the optimized conditions and obtained a 34% isolated yield of **3f** (entry 7, Table 3). (vi) We prepared cyclic glucosides **4** by deacetylation[9] of **3** in 90% to 97% yields (Table 3).

Table 3 Synthesis of cyclic glucosides **3**[a] and **4**



Entry	R	Isolated yields of **3** (%)	Isolated yields of **4** (%)
1	Ph (**2a**)	36 (**3a**)	97 (**4a**)
2	*p*-MeC_6H_4 (**2b**)	38 (**3b**)	92 (**4b**)
3	*p*-$MeOC_6H_4$ (**2c**)	23 (**3c**)	95 (**4c**)
4	*p*-FC_6H_4 (**2d**)	34 (**3d**)	91 (**4d**)
5	*p*-ClC_6H_4 (**2e**)	38 (**3e**)	96 (**4e**)
6	*p*-BrC_6H_4 (**2f**)	31 (**3f**)	90 (**4f**)
7[b]	*p*-BrC_6H_4 (**2f**)	34 (**3f**)	—
8	*p*-IC_6H_4 (**2g**)	24 (**3g**)	92 (**4g**)
9	*p*-$MeOOCC_6H_4$ (**2h**)	36 (**3h**)	94 (**4h**)
10	*p*-$MeCOC_6H_4$ (**2i**)	34 (**3i**)	92 (**4i**)
11	2-thienyl (**2j**)	34 (**3j**)	92 (**4j**)
12	 (**2k**)	20 (**3k**)	91 (**4k**)

[a] All reactions were carried out on a 1.0 mmol scale under the standard conditions: **1** (0.025 M), **2** (1.2 equiv.), $Pd_2(dba)_3$·$CHCl_3$ (2.5 mol%), TFP (10 mol%), K_2CO_3 (2.0 equiv.), DMF–H_2O (80/1), 100 °C, 5 h. [b] The reaction was conducted on a 4.0 mmol scale of **1**.

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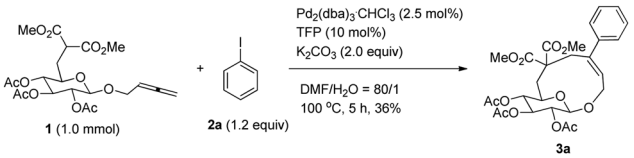
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Conclusions

In conclusion, we have developed a highly regio- and stereoselective synthesis of cyclic β-D-glucosides *via* Pd(0)-catalyzed coupling cyclization of functionalized allenyl β-D-glucoside and organic iodides. Studies on the SGLT2 inhibition activity are in progress. Further study in this area is being pursued in our laboratory.

Experimental

Pd(0)-catalyzed coupling-cyclization of 1 with different organic iodides 2 under standard conditions. Preparation of 3a.



Typical procedure I. To a flame-dried three-necked flask (100 mL) equipped with a reflux condenser containing K_2CO_3 (278.1 mg, 2.0 mmol) were added $Pd_2(dba)_3{\cdot}CHCl_3$ (25.5 mg, 0.025 mmol), TFP (23.5 mg, 0.1 mmol), **1** (477.6 mg, 1.0 mmol), **2a** (251.2 mg, 1.2 mmol), and DMF–H_2O (v/v = 80/1, 40.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 100 °C for 5 h, as monitored by TLC (eluent: petroleum ether–ethyl acetate = 2.5/1). After evaporation of the solvent, the residue was diluted with 60 mL of EtOAc, washed with water and brine, and then dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, chromatography on silica gel (eluent: petroleum ether–ethyl acetate = 2.5/1) afforded **3a** (197.4 mg, 36%) as a solid: M.P. 185–186 °C (*n*-hexane–DCM); $[\alpha]_D^{20}$ = +111.4 (*c* = 0.475, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.32–7.19 (m, 5H, ArH), 5.99 (t, *J* = 8.1 Hz, 1H, CH═), 5.22–5.08 (m, 2H), 5.03–4.93 (m, 1H), 4.86 (s, 1H), 4.41–4.27 (m, 2H), 3.88 (d, *J* = 13.8 Hz, 1H), 3.83–3.67 (m, 1H), 3.74 (s, 3H, Me), 3.29 (d, *J* = 13.8 Hz, 1H), 2.89 (s, 3H, Me), 2.70 (dd, J_1 = 14.7 Hz, J_2 = 12.3 Hz, 1H), 2.13–1.96 (m, 1H), 2.08 (s, 3H, Me), 2.06 (s, 3H, Me), 2.04 (s, 3H, Me); ^{13}C NMR (75 Hz, $CDCl_3$) δ 170.8, 170.4, 169.7, 169.5, 169.3, 145.0, 141.4, 128.2, 127.8, 127.5, 127.0, 102.0, 73.7, 70.8, 69.7, 64.5, 55.6, 52.5, 51.7, 33.6, 32.9, 20.6, 20.5; IR (KBr) ν (cm^{-1}) 2957, 1754, 1730, 1439, 1370, 1243, 1221, 1117, 1080, 1067, 1041; MS (ESI, *m*/*z*) 1119 (2M + Na^+), 587 (M + K^+), 571 (M + Na^+), 566 (M + NH_4^+), 549 (M + H^+); Anal. Calcd for $C_{27}H_{32}O_{12}$ (%): C 59.12, H 5.88; Found: C 59.13, H 5.92.

Acknowledgements

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Notes and references

1 (*a*) N. J. Wareham and N. G. Forouhi, *Diabetologia*, 2005, **48**, 1454; (*b*) S. Wild, G. Roglic, A. Green, R. Sicree and H. King, *Diabetes Care*, 2004, **27**, 1047; (*c*) S. H. Golden, K. A. Robinson, I. Saldanha, B. Anton and P. W. Ladenson, *J. Clin. Endocrinol. Metab.*, 2009, **94**, 1853.

2 (*a*) F. W. Mckee and W. B. Hawkins, *Physiol. Rev.*, 1945, **25**, 255; (*b*) G. Toggenburger, M. Kessler and G. Semenza, *Biochim. Biophys. Acta*, 1982, **688**, 557.

3 (*a*) H. Kakinuma, T. Oi, Y. Hashimoto-Tsuchiya, M. Arai, Y. Kawakita, Y. Fukasawa, I. Iida, N. Hagima, H. Takeuchi, Y. Chino, J. Asami, L. Okumura-Kitajima, F. Io, D. Yamamoto, N. Miyata, T. Takahashi, S. Uchida and K. Yamamoto, *J. Med. Chem.*, 2010, **53**, 3247; (*b*) R. P. Robinson, V. Mascitti, C. M. Boustany-Kari, C. L. Carr, P. M. Foley, E. Kimoto, M. T. Leininger, A. Lowe, M. K. Klenotic, J. I. MacDonald, R. J. Maguire, V. M. Masterson, T. S. Maurer, Z. Miao, J. D. Patel, C. Préville, M. R. Reese, L. She, C. M. Steppan, B. A. Thuma and T. Zhu, *Bioorg. Med. Chem. Lett.*, 2010, **20**, 1569; (*c*) S. Nomura, S. Sakamaki, M. Hongu, E. Kawanishi, Y. Koga, T. Sakamoto, Y. Yamamoto, K. Ueta, H. Kimata, K. Nakayama and M. Tsuda-Tsukimoto, *J. Med. Chem.*, 2010, **53**, 6355; (*d*) W. Meng, B. A. Ellsworth, A. A. Nirschl, P. J. McCann, M. Patel, R. N. Girotra, G. Wu, P. M. Sher, E. P. Morrison, S. A. Biller, R. Zahler, P. P. Deshpande, A. Pullockaran, D. L. Hagan, N. Morgan, J. R. Taylor, M. T. Obermeier, W. G. Humphreys, A. Khanna, L. Discenza, J. G. Robertson, A. Wang, S. Han, J. R. Wetterau, E. B. Janovitz, O. P. Flint, J. M. Whaley and W. N. Washburn, *J. Med. Chem.*, 2008, **51**, 1145; (*e*) B. Lv, B. Xu, Y. Feng, K. Peng, G. Xu, J. Du, L. Zhang, W. Zhang, T. Zhang, L. Zhu, H. Ding, Z. Sheng, A. Welihinda, B. Seed and Y. Chen, *Bioorg. Med. Chem. Lett.*, 2009, **19**, 6877; (*f*) N. C. Goodwin, R. Mabon, B. A. Harrison, M. K. Shadoan, Z. Y. Almstead, Y. Xie, J. Healy, L. M. Buhring, C. M. DaCosta, J. Bardenhagen, F. Mseeh, Q. Liu, A. Nouraldeen, A. G. E. Wilson, S. D. Kimball, D. R. Powell and D. B. Rawlins, *J. Med. Chem.*, 2009, **52**, 6201; (*g*) B. Xu, Y. Feng, B. Lv, G. Xu, L. Zhang, J. Du, K. Peng, M. Xu, J. Dong, W. Zhang, T. Zhang, L. Zhu, H. Ding, Z. Sheng, A. Welihinda, B. Seed and Y. Chen, *Bioorg. Med. Chem.*, 2010, **18**, 4422; (*h*) H. Zhou, D. P. Danger, S. T. Dock, L. Hawley, S. G. Roller, C. D. Smith and A. L. Handlon, *ACS Med. Chem. Lett.*, 2010, **1**, 19; (*i*) S. Y. Kang, K. Song, J. Lee, S. Lee and J. Lee, *Bioorg. Med. Chem.*, 2010, **18**, 6069.

4 (*a*) V. Mascitti and C. Préville, *Org. Lett.*, 2010, **12**, 2940; (*b*) S. Y. Kang, M. J. Kim, J. S. Lee and J. Lee, *Bioorg. Med. Chem.*, 2011, **21**, 3759.

5 (*a*) X. Jiang, Q. Yang, Y. Yu, C. Fu and S. Ma, *Chem. – Eur. J.*, 2009, **15**, 7283; (*b*) B. Wan, G. Jia and S. Ma, *Adv. Synth. Catal.*, 2011, **353**, 1763; (*c*) X. Huang, X. Jiang, C. Fu and S. Ma, *Adv. Synth. Catal.*, 2013, **355**, 3295.

6 (*a*) G. Illuminati and L. Mandolini, *Acc. Chem. Res.*, 1981, **14**, 95; (*b*) G. A. Molander, *Acc. Chem. Res.*, 1998, **31**, 603.

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7 For the synthesis of allenyl β-D-glucoside **1**, see: X. Huang, C. Fu and S. Ma, *Synthesis*, 2014, 2917.

8 Crystal data for compound **5**: $C_{42}H_{38}O_{12}$, M_W = 734.72, monoclinic, space group *P*21, final *R* indices [$I > 2\sigma(I)$], R_1 = 0.0354, wR_2 = 0.0945, *R* indices (all data) R_1 = 0.0383, wR_2 = 0.0981, a = 7.29191(19) Å, b = 16.5352(4) Å, c = 16.0030(4) Å, α = 90.00°, β = 99.499(3)°, γ = 90.00°, V = 1903.08(8) Å^3, T = 293(2) K, Z = 2, reflections collected/unique 7884/3510 (R_{int} = 0.0336), number of observations [>2$\sigma(I)$] 3260, parameters: 490. CCDC-987494 contains the supplementary crystallographic data for this paper.

9 N. Pietrzik, D. Schmollinger and T. Ziegler, *Beilstein J. Org. Chem.*, 2008, **4**, 30.

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